UNIVERSITY OF MANAGEMENT AND TECHNOLOGY, SIALKOT

Design and Analysis of Algorithms (CC342)

Assignment 1 - Foundations of Algorithm Analysis

Student Name		Registration No.	
Section		Total Marks	50

Course Learning Outcomes (CLOs) Assessed

CLO	Statement
CLO1	Explain what is meant by “best”, “expected”, and “worst” case behavior of an algorithm
CLO2	Identify the characteristics of data and/or other conditions or assumptions that lead to different behaviors.
CLO3	Determine informally the time and space complexity of simple algorithms
CLO4	List and contrast standard complexity classes
CLO5	Use big O, Omega, Theta notation formally to give asymptotic upper bounds on time and space complexity of algorithms

Question–CLO Mapping

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
CLO1, CLO2	CLO3	CLO3	CLO3	CLO2, CLO3	CLO4	CLO5	CLO2, CLO5

Coverage & Instructions

• Coverage: Lectures 1–4 (Weeks 1–2): algorithmic thinking, Fibonacci, GCD, runtime analysis, growth rates, Big-O, Ω, Θ, and computational-model assumptions.

• Show essential working for dry runs and asymptotic calculations. Answers without reasoning may receive partial credit.

• Unless a question states otherwise, use the simplified unit-cost model for basic arithmetic operations.

• For runtime estimates involving logarithms, use base 2.

Question 1. Algorithmic Thinking and Case Analysis **[5 marks]** *(CLO1, CLO2)*

Answer briefly.

(a) Differentiate a straightforward programming problem from an algorithmic problem. Give one example of each from the lectures. [2]

(b) Why can exact execution time in seconds be a poor way to compare algorithms across computers? [1]

(c) For linear search, state the best-case, expected-case, and worst-case asymptotic runtimes. Assume the target is equally likely to occur at any array position when discussing the expected case. Also identify when the best and worst cases occur. [2]

Question 2. Fibonacci: Definition and Algorithms [7 marks] (CLO3)

Use the Fibonacci sequence defined by:

$$F_0 = 0, \quad F_1 = 1, \quad \text{and} \quad F_n = F_{n-1} + F_{n-2}$$

(a) Write F_2 through F_8. [2]

(b) Explain in one or two sentences why the naive recursive Fibonacci algorithm is slow. [2]

(c) Dry-run FibList(6) by showing the values stored in F[0] through F[6]. [2]

(d) State the asymptotic space complexity of FibList(n) under the array-based implementation and briefly justify it. [1]

Question 3. Counting FibList Steps [5 marks] *(CLO3)*

Under the simple line-counting model used in the lecture:

$$T(n) = 2n + 2$$

(a) Explain why the loop from i = 2 to n executes $n - 1$ times. [2]

(b) Compute T(10) and T(100). [2]

(c) State the Big-O growth of T(n) = 2n + 2. [1]

Question 4. Greatest Common Divisor [8 marks] *(CLO3)*

Consider gcd(48, 18).

(a) List the common factors of 48 and 18 and state the GCD. [2]

(b) Dry-run the Euclidean Algorithm, showing each pair (a, b) until the base case is reached. [4]

(c) State the key lemma used by the Euclidean Algorithm in the form gcd(a, b) = ... [2]

Question 5. Naive GCD versus Euclidean GCD [5 marks] *(CLO2, CLO3)*

Consider the lecture version of NaiveGCD, whose loop runs from d = 1 to a + b.

(a) For a = 18 and b = 12, how many loop iterations are performed? [1]

(b) Why is this deliberately inefficient? Give a tighter brute-force stopping point for positive a and b. [2]

(c) In one sentence, explain why EuclidGCD is much more efficient conceptually. [2]

Question 6. Growth Rates and Scalability **[7 marks]** *(CLO4)*

(a) Arrange the following from slowest to fastest asymptotic growth: [3]

$$n^2,\quad \log n,\quad 2^n,\quad n,\quad \sqrt{n},\quad n \log n$$

(b) Assume 10^9 basic operations per second and $n = 10^6$. Estimate the time for an $n \log_2 n$ algorithm and an n^2 algorithm. [3]

(c) Which of the two scales better, and why? [1]

Question 7. Big-O and Dominant Terms **[8 marks]** *(CLO5)*

Simplify each expression to its most informative standard Big-O bound.

(a) $7n + 20$ [1]

(b) $3n^2 + 5n + 2$ [1]

(c) $4n \log_2 n + 7$ [1]

(d) $2^n + n^9$ [1]

(e) Using the formal definition of Big-O, show that $3n^2 + 5n + 2 = O(n^2)$ by giving suitable constants c and N. [4]

Question 8. Asymptotic Notation and Computational Model **[5 marks]** *(CLO2, CLO5)*

(a) Complete the statements: Big-O is an asymptotic _____ bound; Big-Ω is a _____ bound; Big-Θ is a _____ bound. [1]

(b) Show that $3n^2 + 5n + 2 = \Theta(n^2)$ by explaining why both $O(n^2)$ and $\Omega(n^2)$ bounds hold. [2]

(c) Why can FibList be described as $O(n)$ under unit-cost arithmetic but $\Theta(n^2)$ under bit-complexity analysis? [2]

End of Assignment • Total: 50 marks